

Pathogenesis of the anemia of uremia: Role of secondary hyperparathyroidism

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Patients with chronic renal failure almost always have marked anemia [1-5]. The various factors that may underlie the anemia of uremia are listed in Table 1. Decreased production [3, 5, 6] and shortened survival of erythrocytes (RBC) [7-11] have been implicated as major causes of anemia in patients with advanced renal failure.

Reduced availability of erythropoietin [12-16] and the presence of inhibitor(s) to various stages of erythropoiesis [5] could be responsible for the decrement in RBC production. Indeed, sera samples from uremic patients have been found to inhibit erythroid colony formation in vitro [17]. Although the nature of these inhibitors is, as yet, not fully elucidated, the polyamine spermine, which is elevated in the blood of uremic patients [18], has been identified as one of these inhibitors [19].

The shortened survival of RBCs in patients with chronic uremia is due to extracorporeal factors, because RBCs from uremic patients have normal survival when infused into other uremic patients, whereas RBCs from normal individuals display shortened survival when infused into uremic patients [10, 20]. Using ^{51}Cr and di (isopropofluoro) ^{32}P techniques, evaluation of RBC survival in patients treated with maintenance hemodialysis revealed that the half-life of the RBCs is markedly reduced [21] and that increasing the frequency of dialysis did not normalize the half-life of the cells [22]. These observations indicate that the serum factor(s) responsible for the reduced survival of the RBC is not effectively removed by dialysis.

Although the anemia of uremia is normochromic normocytic in nature, certain deficiency states may develop in these patients and may worsen the anemia and alter its nature. Excessive blood losses are not uncommon in uremic patients and may lead to a state of iron deficiency. Blood loss may occur in the dialysis machine, may be secondary to a hemostatic defect and increased bleeding tendencies [23-25], or be due to frequent blood sampling for the monitoring of the patients. Folic acid deficiency resulting in megaloblastic anemia may complicate the anemia of uremia in dialysis patients [26]. The folic acid deficiency is due to its loss during the dialysis procedure [27] and to inadequate intake.

A prominent biochemical abnormality in renal failure is secondary hyperparathyroidism [28-30] leading to marked elevation in the blood levels of parathyroid hormone (PTH) [31-33]. Recently, PTH has been implicated to be a major uremic toxin underlying, at least partly, many of the manifestations of the uremic syndrome [34-36]. It is possible that the excess blood levels of PTH play an important role in the pathogenesis of the anemia of uremia. Several clinical observations support this contention.

Avram et al [37] found in 12 dialysis patients that their hematocrit and reticulocytes increased and their blood transfusion requirements decreased after subtotal parathyroidectomy. Better et al [38] and Shasha et al [39] reported a significant rise in hemoglobin in four uremic patients after parathyroidectomy, and they mentioned one patient in whom the hemoglobin increased from 4.0 to 9.0 g/dl after suppression of the activity of the parathyroid glands by treatment with 1,25 dihydroxycholecalciferol. Similar observations were reported by Zingraff et al [40] and Barbour [41] in many of their patients who underwent parathyroidectomy. Anemia has also been reported in patients with primary hyperparathyroidism. Aurbach et al [42] and Mallette et al [43] found that 12 of 58 (21%) patients with parathyroid adenoma had anemia that could not be explained by other causes; in 7 of 8 such patients, the anemia disappeared after removal of the adenoma. Also, Boxer et al [44] reported that 17 of their 322 (5.1%) patients with primary hyperparathyroidism had unexplained normochromic normocytic anemia. They also found that the blood levels of PTH were four times higher in the patients with anemia than in those without anemia. In 7 of their patients in whom data were available before and after removal of the parathyroid adenoma, there were spontaneous and significant ($P < 0.01$) increments in hematocrit from $29.6 \pm (\text{SEM}) 1.3$ to $40.1 \pm 1.5\%$.

Several investigators studied the interaction between PTH and the hematopoietic system and provided information on the pathways through which PTH may induce anemia. Studies in our laboratories [45] demonstrated that intact PTH (1-84 PTH) in concentrations (7.5 to 30 U/ml) comparable to those found in the blood of uremic patients produced marked inhibition of human peripheral blood and mouse bone marrow erythroid

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Table 1. Factors that may underlie the anemia of uremia

Decreased erythropoiesis:
Reduced availability of erythropoietin
Inhibitor(s) of erythropoiesis
Bone marrow fibrosis
Shortened red blood cell survival:
Hemolysis due to extracorporeal factor(s)
Excessive blood losses
Deficiency states:
Iron deficiency
Folic acid deficiency

burst-forming units (BFU-E) but had no effect on mouse bone marrow erythroid colony-forming units (CFU-E). Inactivation of 1-84 PTH abolished its action on erythropoiesis. Increasing the concentration of erythropoietin in the media from 0.07 to 1.7 U/ml overcame the inhibitory effect of 1-84 PTH on BFU-E. The NH₂-terminal fragment of PTH (1-34 PTH) and the COOH-terminal fragment of PTH (53-84 PTH) had no effect on BFU-E. These data demonstrate that either the intact PTH or a COOH-terminal fragment(s) bigger than the 53-84 moiety exerts the inhibitory effect on erythropoiesis, and adequate amounts of erythropoietin can overcome this action.

The demonstration that erythropoietin can overcome the inhibitory effect of PTH on erythropoiesis is interesting. It provides an explanation for the lower prevalence and the milder degree of anemia in patients with primary hyperparathyroidism. These patients usually have normal kidneys and can produce adequate amounts of erythropoietin, which would counterbalance the effect of PTH on erythropoiesis. In contrast, patients with chronic renal failure are in double jeopardy; the excess blood levels of PTH inhibit erythropoiesis, and the inability to generate adequate erythropoietin permits the inhibitory action of PTH to proceed unchecked. The interaction between erythropoietin and PTH also provides the experimental basis for the understanding of the clinical observations reported by Caro et al [16]. These authors studied two groups of uremic patients with similar levels of hematocrit. One group had significantly higher blood levels of PTH and erythropoietin as well. Their data would suggest that the inhibitory effect of PTH on erythropoiesis in those with higher blood levels of hormone was overcome by the higher blood levels of erythropoietin, resulting in hematocrit levels similar to those noted in the patients with lower PTH levels and, hence, lesser inhibition of erythropoiesis.

As mentioned before, spermine also inhibits erythropoiesis [19], but its inhibitory effect occurs at the CFU-E stage. Thus, at least two inhibitors of erythropoiesis in uremic patients have been well documented: PTH acts at the BFU-E level, and spermine acts at the CFU-E level. Spermine is dialyzable and, therefore, dialysis therapy may ameliorate its inhibitory effect.

Other investigators [46] reported that high concentrations of PTH inhibit RNA and heme synthesis by mouse embryonic liver erythroid precursors. Also RBC production, as measured by ⁵⁹Fe incorporation, is reduced in rats with excess PTH as compared to control rats [47]. These observations provide further support for a role of the secondary hyperparathyroidism

in the genesis of the anemia of uremia. Finally, Dunn and Trent [48] challenged this notion, because they were able to demonstrate inhibition of heme synthesis only very high concentration of PTH (240 times normal). They indicated that such concentrations are higher than those observed in the blood of uremic patients.

Studies in our laboratories have also demonstrated that PTH may be an important factor responsible for the shortened RBC survival [48]. We found that both the 1-34 PTH and 1-84 PTH, but not the COOH-terminal (53-84 PTH), produced significant increases in osmotic fragility of human RBCs. This effect was abolished by prior activation of the hormone. There was a dose-response relationship between both moieties of PTH and the increase in osmotic fragility. This action of PTH required calcium, was mimicked by calcium ionophore, and was partially blocked by verapamil. PTH caused significant influx of ⁴⁵Ca into RBCs. Scanning electron microscopy revealed that the incubation of RBCs with PTH was associated with the appearance of membrane filamentous extensions which anchor RBCs together. PTH did not affect water content of RBC. Inhibition of glycolytic activity of RBCs with NaF or inhibition of sodium-potassium-activated ATPase with ouabain did not abolish the effect of PTH on osmotic fragility. PTH did not stimulate the sodium-potassium-activated ATPase or the magnesium-dependent ATPase, but caused marked and significant stimulation of calcium-activated ATPase. The basal activity of RBC adenylate cyclase was low, and PTH produced only a modest stimulation of this enzyme. Both cyclic AMP and dibutyryl cyclic AMP had no effect on osmotic fragility. These observations indicate that: (1) the human RBC is a target organ for PTH, (2) the hormone increases osmotic fragility of the RBC, and (3) this effect of PTH is due to enhanced calcium entry into RBCs. It is possible that the increased calcium influx may affect the spectrin-action of the cytoskeletal network of the RBC and alter the stability and integrity of the cell membrane.

Excess PTH induces moderate to marked degrees of bone marrow fibrosis [49, 50] and, therefore, limits the availability of red cell-forming units. At least three groups of investigators have reported a relationship between the degree of bone marrow fibrosis and anemia in patients with renal failure [40, 51, 52]. Zingraff et al [40] documented such a relationship in uremic patients studied before and after parathyroidectomy. Weinberg et al [51] described myelofibrosis in dialysis patients and documented its relation to their anemia. Wainer et al [52] reported higher prevalence of endosteal fibrosis (5 of 9 patients, 55%) in uremic patients with hemoglobin of less than 8.0 g/dl than in those with higher hemoglobin (2 of 11 patients, 18%).

Summary. PTH may participate in the genesis of the anemia of uremia through at least three pathways. These include inhibition of erythropoiesis, shortening survival of RBCs and inducing fibrosis of bone marrow cavity. A possible fourth mechanism through which PTH may contribute to the anemia of uremia is its effect on platelets. PTH inhibits platelet aggregation [53] and, as such, may play an important role in the genesis of the bleeding tendencies and the consequent blood loss in uremia.

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